Introduction
to
Reinforcement Learning based Control
and
Deep Reinforcement Learning based control

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Introduction

Markov Decision process

Dynamic Programming for planning and control

Model Free Prediction and Control (SARSA and Q-learning)

Function Approximation (Use Deep Neural Networks)

Deep Q networks (deep learning for RL)

Deep Deterministic Policy Gradient Method

Introduction

Environment

Agent

Reward

Policy

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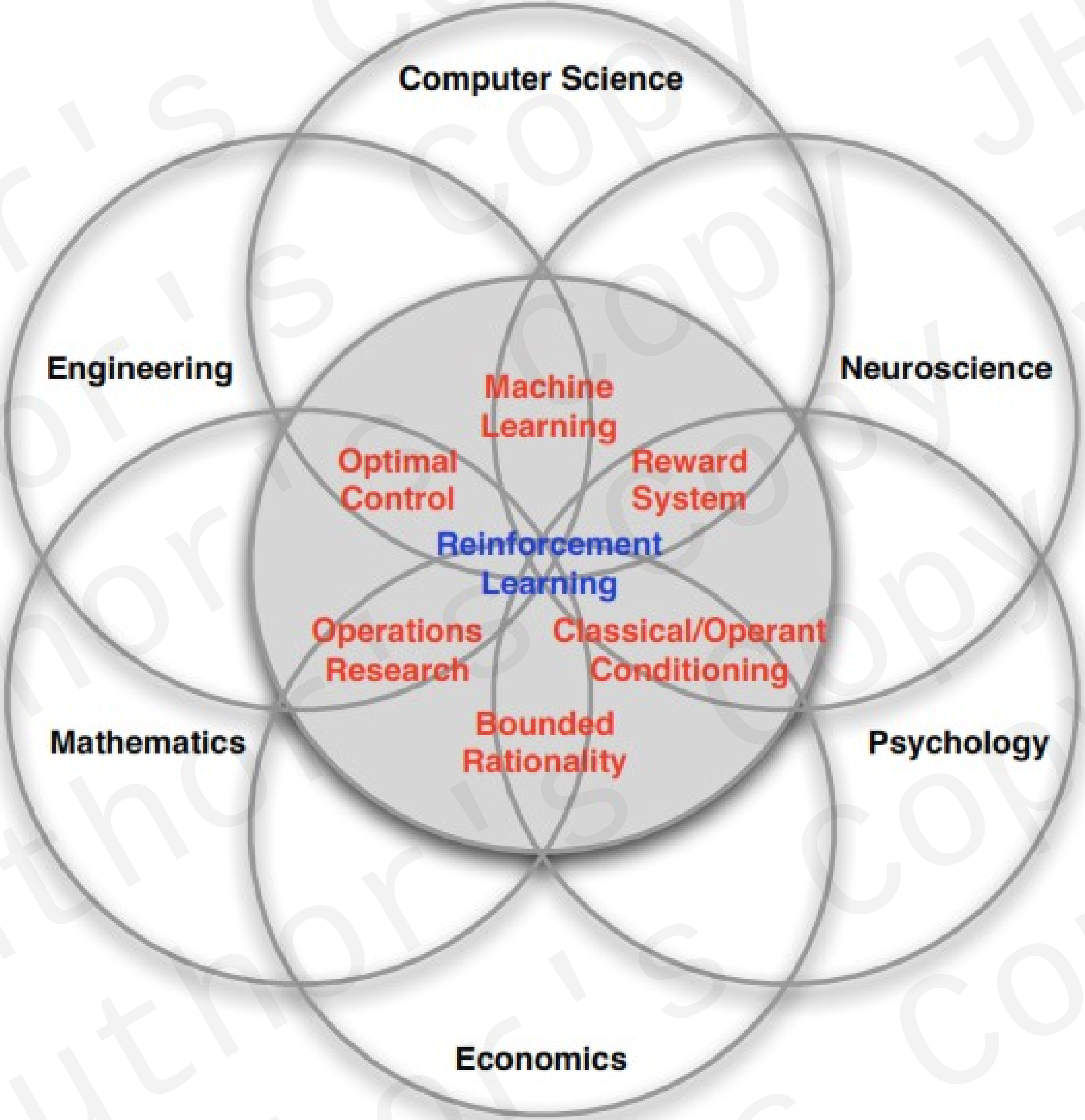
Computer Science
Engineering
Neuroscience
Machine Learning
Optimal Control
Reward System
Reinforcement Learning
Operations Research
Classical/Operant Conditioning
Bounded Rationality
Mathematics
Psychology
Economics

Motivation

Reinforcement Learning: Towards human level :

control ((Finding the optimal way of doing a given task)

prediction

Adaptation (Robots That Can Adapt like Animals, *Nature*)

Built new moves

POLYTECH NANCY

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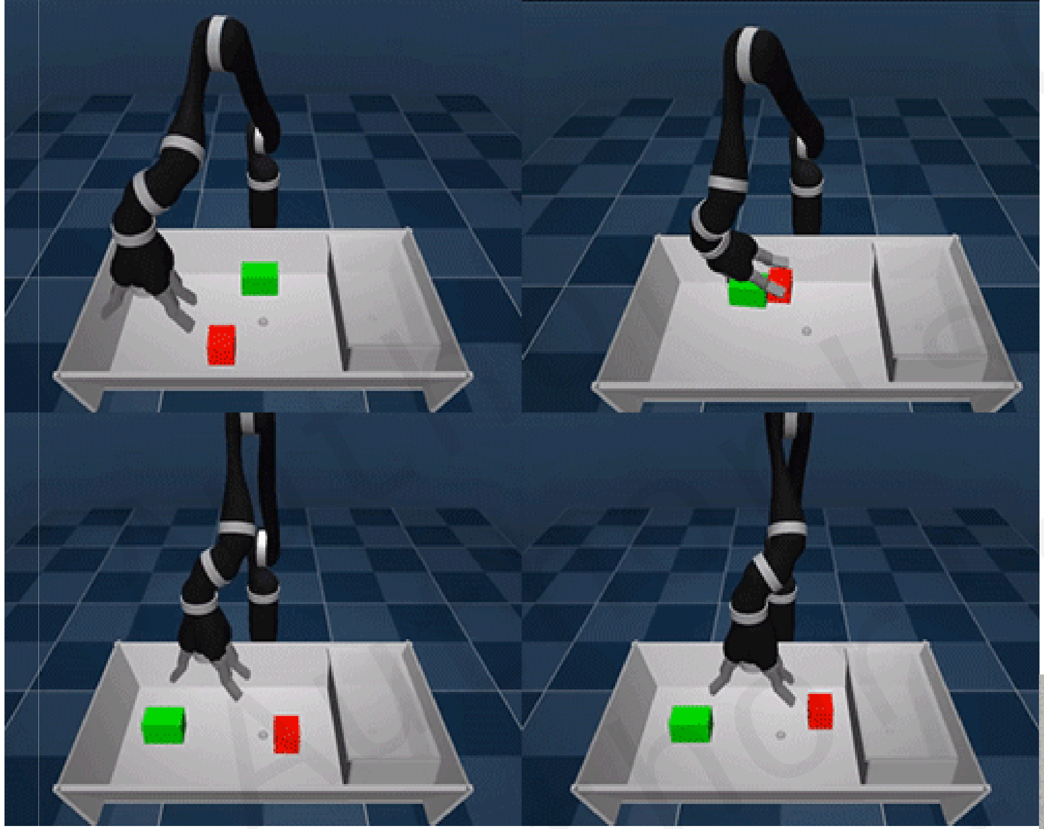
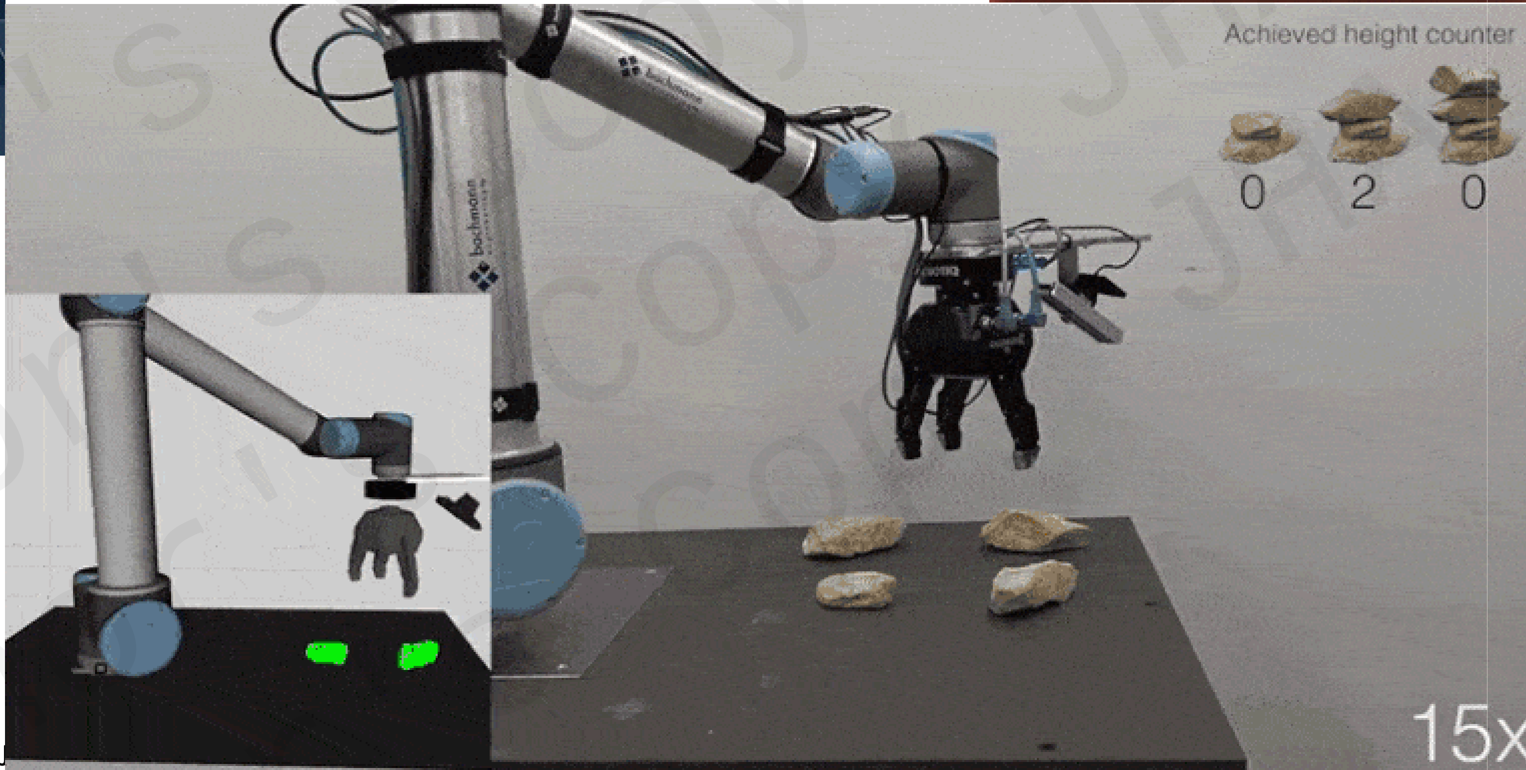


Some Applications

Source : Deep Mind



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Agent

obervation

O_t

Agent/controller

Action

A_t

Reward

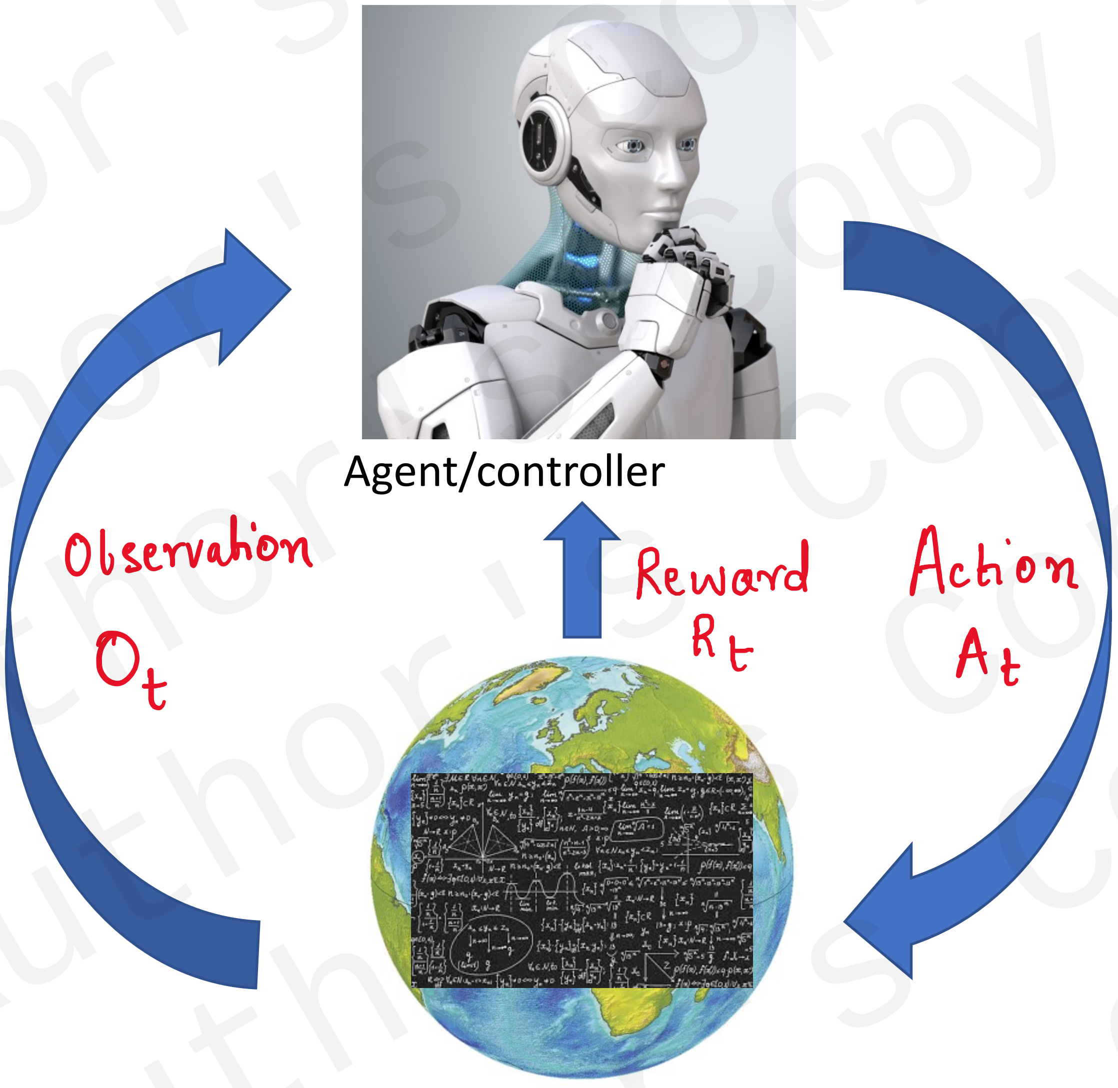
R_t

Agent and Environment

Agent/controller

Environment/system

Agent and Environment and Interaction



Environment/system

At each step t:

Agent:

executes action A_t

Receives observation O_t

Receives reward R_t

Environment:

Receives action A_t

Emits observation O_{t+1}

Emits scalar reward R_{t+1}

$t \leftarrow t+1$

Rewards

Reward R_t : scalar feedback signal

: indicates how well an agent does

: Agent maximises the cumulated reward.

Reinforcement Learning: find the best way (optimal way) to maximize the cumulative reward with respect to a given goal.

Sequential Decision Making.

Goal → Select actions that maximize the total future reward.

To Note
- Immediate actions may have long term consequences.
- Rewards may be delayed.
- Perhaps better to sacrifice an immediate large reward for far larger long term reward.

History

History : Sequences of observations, actions and rewards.

$$H_t = O_1, R_1, A_1, O_2, R_2, A_2 \ldots\ldots A_{t-1}, O_t, R_t$$

(all observable variables till time t)

What Next ??

Depending on history:

- Agent selects actions!
- Environment selects observations / rewards.

History and State

State:
- information used to determine what happens next.
- state is function of history.

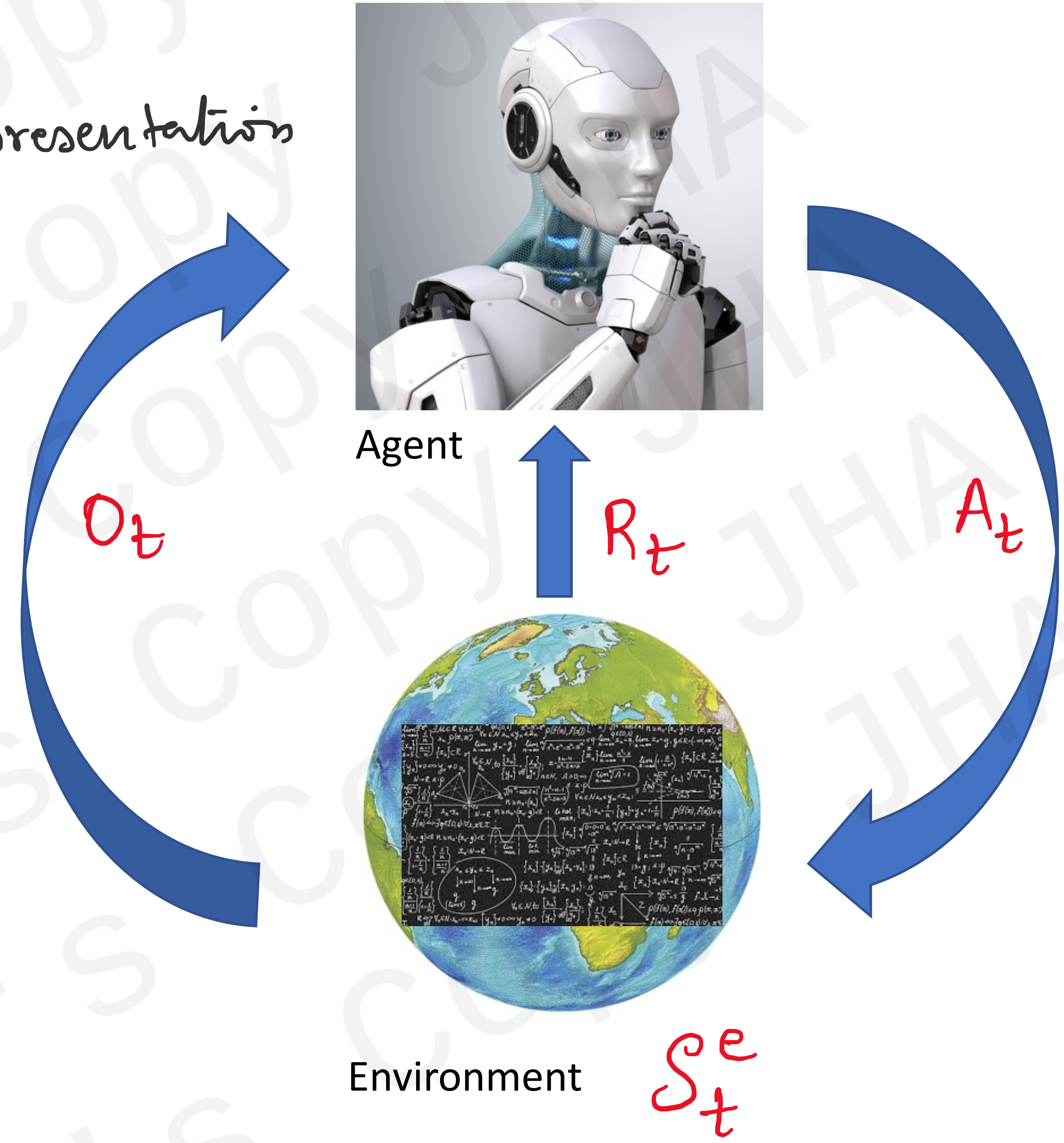
$$S_t = f(H_t)$$

Environment State

S_t^e → environment's privet representation

→ Whatever knowledge env uses to emit next observation or reward

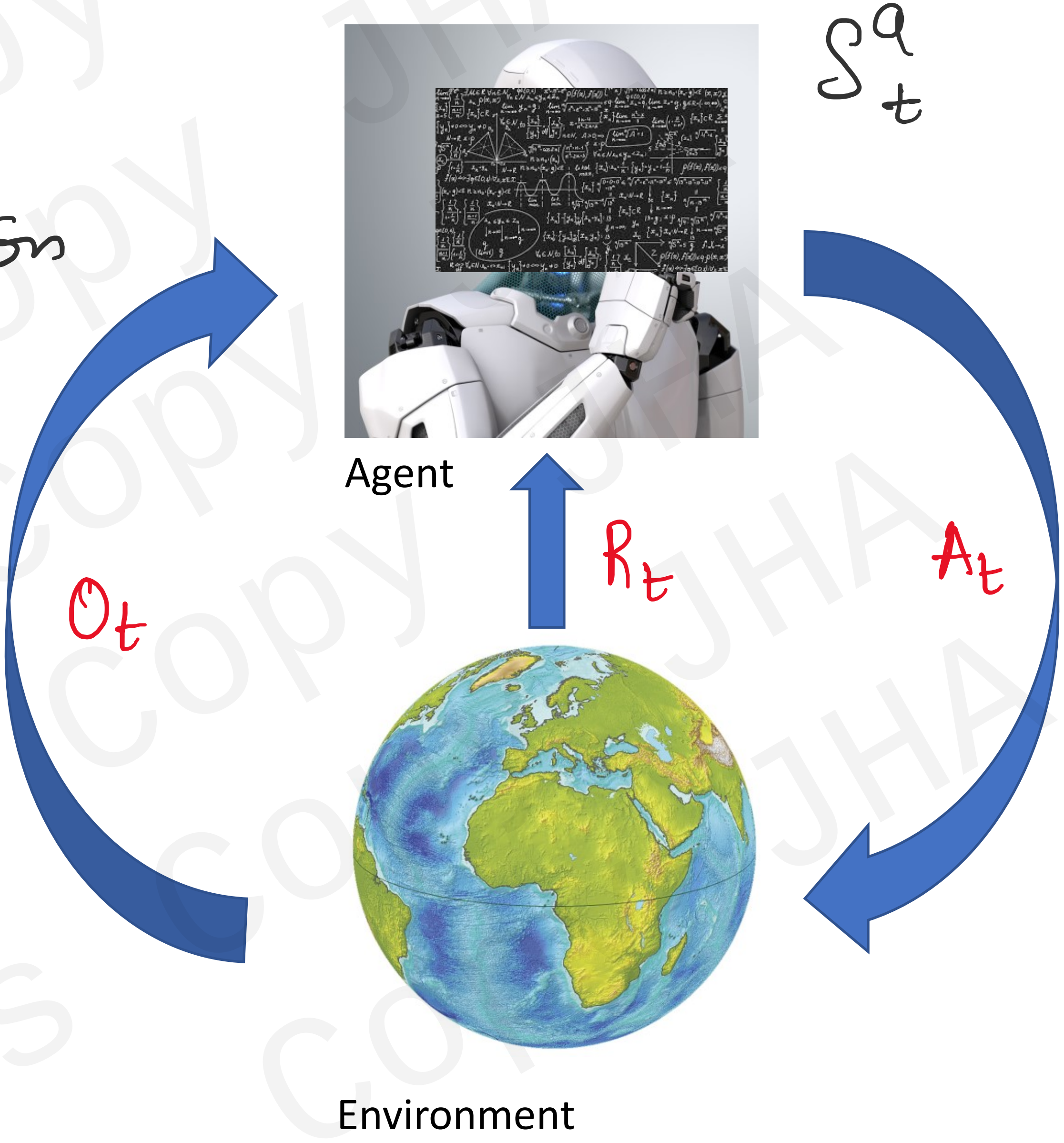
→ State may not be visible



Agent State S_t^a

- $S_t^a \rightarrow$ agent's internal representation
- the essential information to pick next action
- it can also be function of history.

  $S_t^a = f(H_t)$



Information State

State S_t → Markov Property

$$P(S_{t+1} | S_t) = P(S_{t+1} | S_1, S_2, \dots S_t)$$

- The future is independant of the past given the present.

$$H_{1:t} \longrightarrow S_t \longrightarrow H_{t+1:\infty}$$

- History is known, throw away the history!
- State is fully capable of describing the 'present' situation of the system (agent or Environment)
- S_t^e is Markov, H_t is Markov!

Markov Decision Process:

MDP : 4 tuple (S, A, P_a, R_a)

$S \rightarrow$ set of states

$A \rightarrow$ set of actions

$P_a(s, s') = Pr(s_{t+1} = s' \mid s_t = s, a_t = a)$

$R_a(s, s')$ = immediate reward

Diagram

All processes in this course $\rightarrow$ MDP!!

Environments.

Fully observable (in this course)

$$O_t = S_t^a = S_t^e.$$

When environments is not fully observable,
we have Partially observable Markov Process (POMDPs)

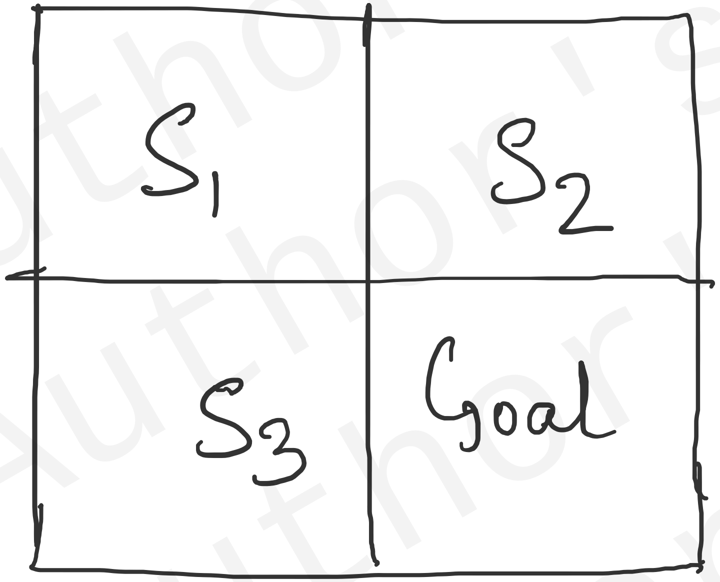
Ex: Robot using camera → position (is not absolute)
Trading agent → observes only prices.
(trends not seen)

Agent:

Agent:

- Policy
- Value Function
- Model.

Grid World Example



What is the environment?

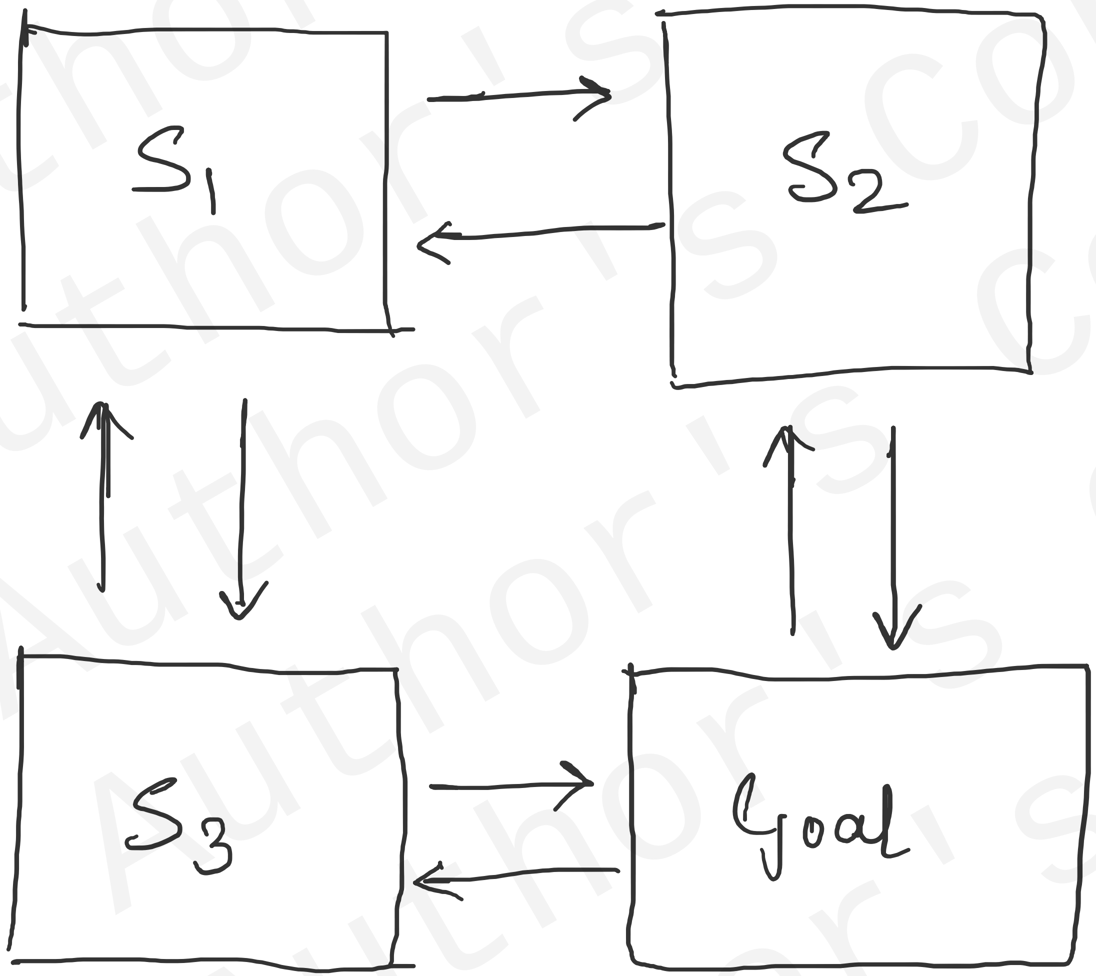
What are the states?

What is agent's role?

A small Grid world $\longrightarrow$ Environment

An agent lives in it $\longrightarrow$ Agent

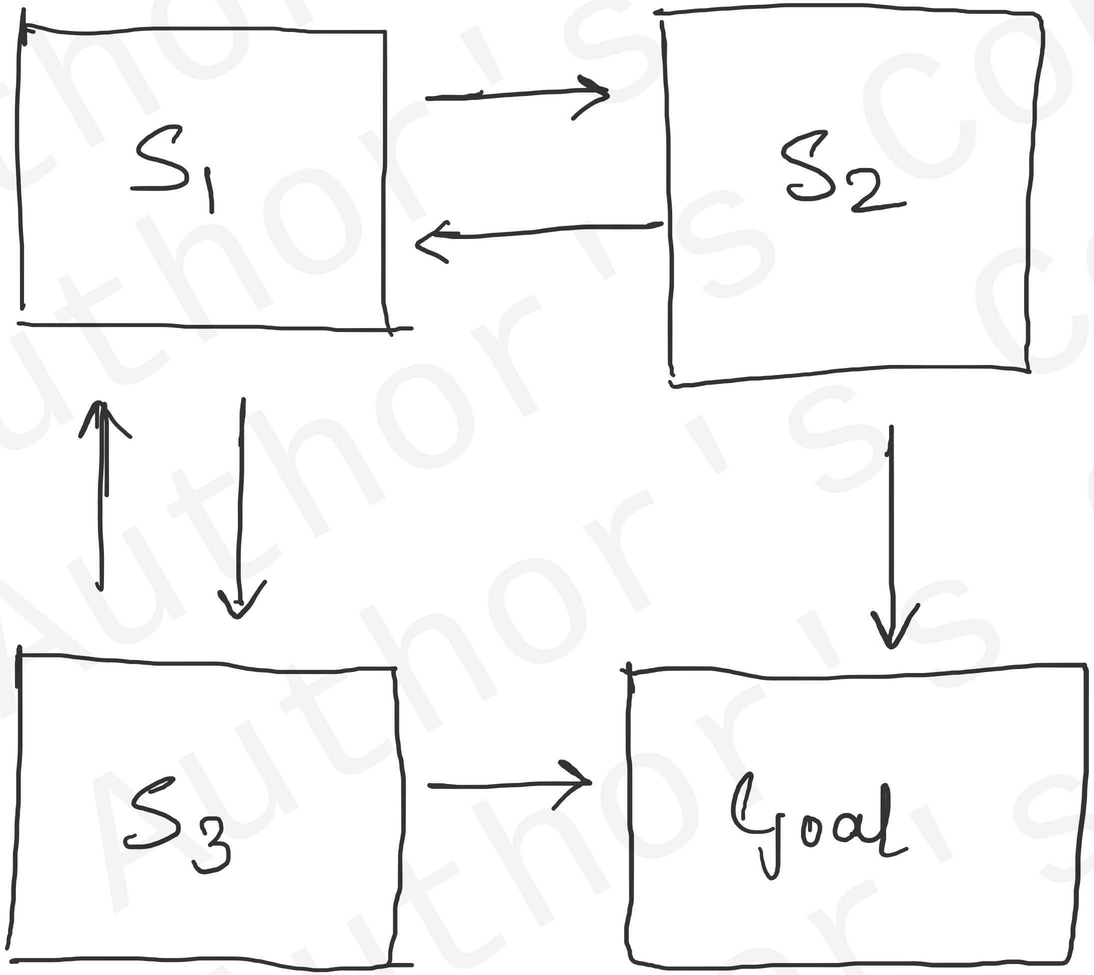
Grid World Example



Agent moves.

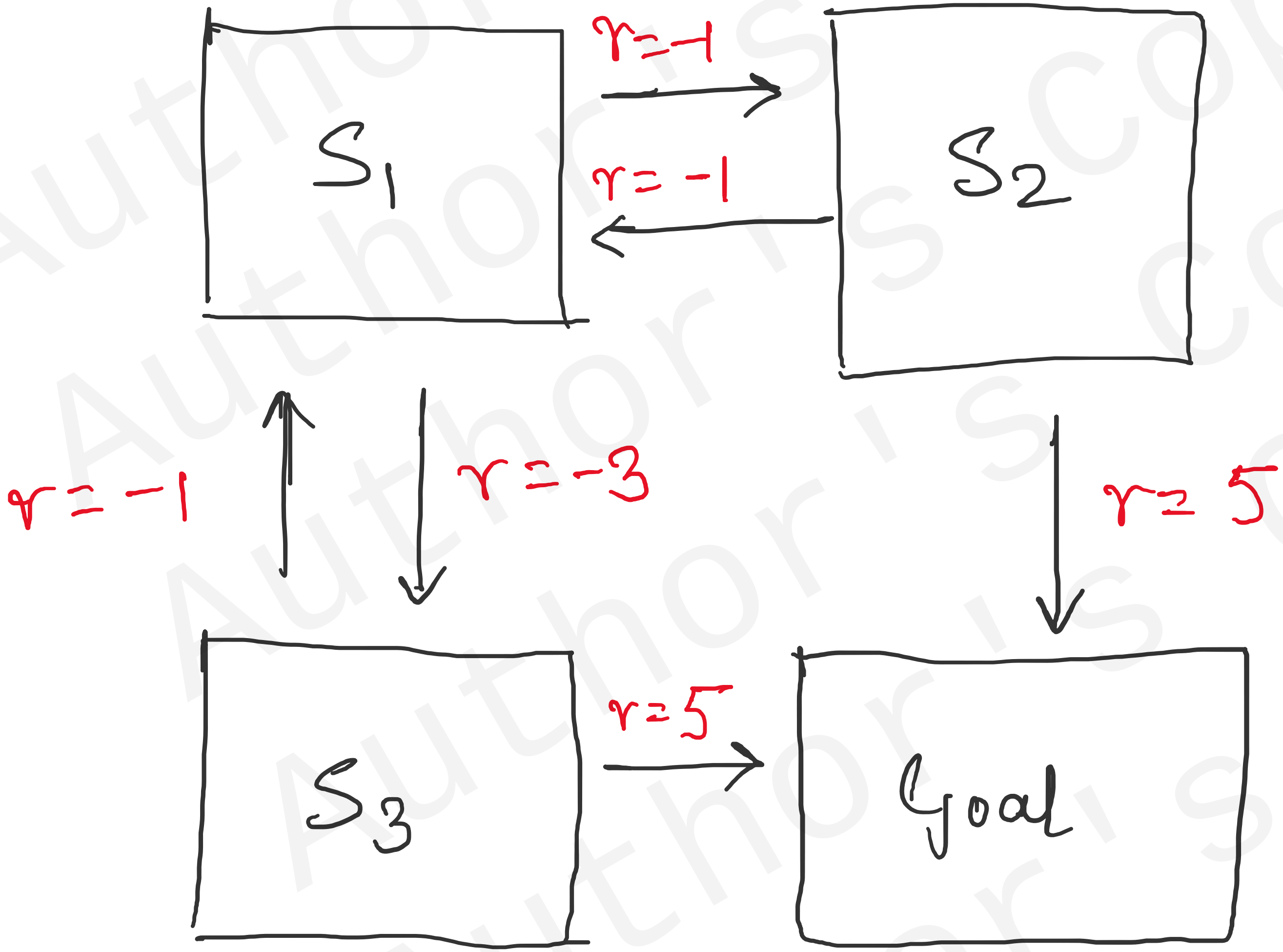
But once reaches goal state, the objective is accomplished!!

Grid World Example



Once, Goal is reached,
No turning back!!

Grid World Example



Rewards are assigned.

Agent knows the MDP

Environment

Rewards

What is the best way to reach goal?

Optimal path?

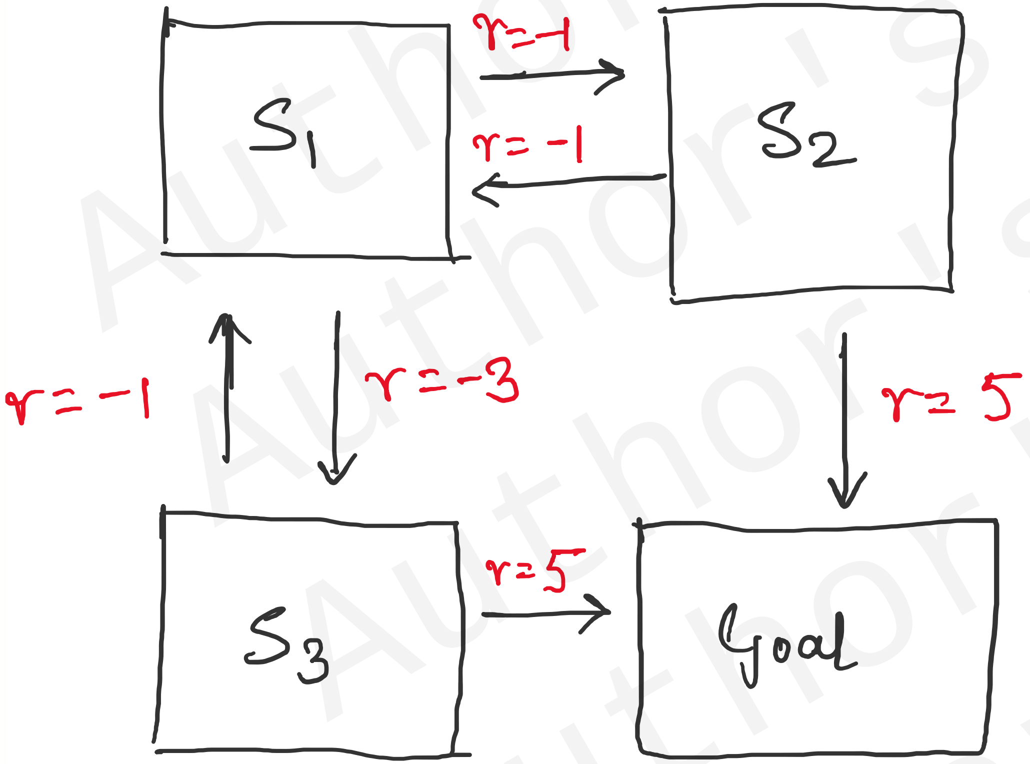
Agent Policy

- Policy → agent's behaviour (dynamics)
- Mathematically, mapping from state to action.

Deterministic Policy = $a = \pi(s)$

Stocastic Policy $\pi(a|s) = P(A_t = a | S_t = s)$

Grid World Example



Policy Example: π

Actions are drawn from a uniform distribution.

$\pi(\text{right} \mid s_1) = 1/2$

$\pi(\text{down} \mid s_1) = 1/2$

$\pi(\text{left} \mid s_2) = 1/2$

$\pi(\text{down} \mid s_2) = 1/2$

$\pi(\text{up} \mid s_2) = 1/2$

$\pi(\text{right} \mid s_2) = 1/2$

Actions are based on policy.

Policy depends on some probability

Agent

Value Function.

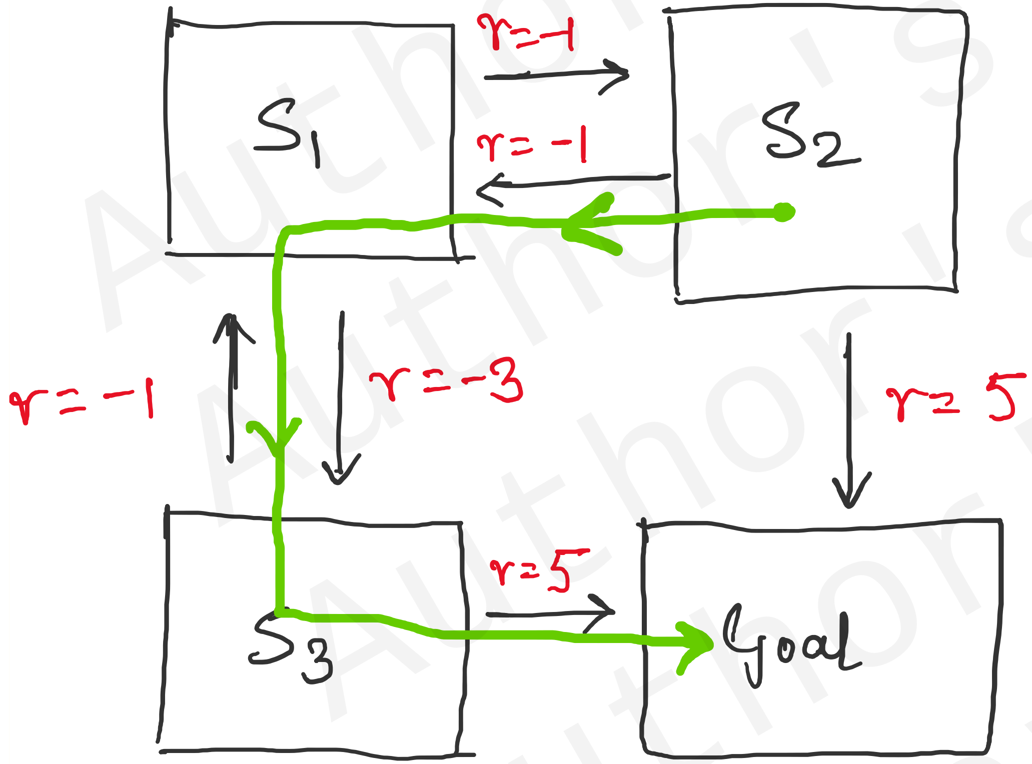
- Assessment of the value of being in a state.
- Prediction of future reward.
- Determines the quality of being in that 'state'.
- Helps to make a choice in taking actions while being in a state.

$$V_\pi(s) = \mathbb{E}_\pi\left[\underbrace{R_{t+1} + R_{t+2} + R_{t+3} + \dots}_{\text{sum of all future rewards.}} \,\middle|\, S_t = s\right]$$

Value of being in 's' with respect to policy π

Mean over policy π

Grid World Example



$$V_\pi(s) = \mathbb{E}_\pi\left[\underbrace{R_{t+1} + R_{t+2} + R_{t+3} + \cdots}\ \middle|\ S_t = s\right]$$

Assume : A movement from S2 as:

$$V_\pi(S_{2,t}) = \mathbb{E}_\pi\left(-1 + -3 + 5 \ \middle|\ S_t = S_2\right)$$

$$= (-1 + -3 + 5) = 1$$

→ we consider only one π

if, $\pi = \{\pi_1, \pi_2, \pi_3 \ldots\}$

then average over all π.

Agent:

Value Function

$$V_\pi(s) = \mathbb{E}_\pi\left[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} \cdots \mid S_t = s\right]$$

Discount factor.

Determines how important is any future reward wrt the present reward.

(if $\gamma = 0.1$; then $\gamma^{10} = (0.1)^{10}$

$\gamma^{10} \ll \gamma$

so $\gamma^{10} \cdot R_{t+11}$ is very less important!!

Model

- Model predicts what environment does next.

  Ex: If you have model of robot → predict next step.

  But exact models are rarely available.

  Exact models do not exist!!

- $P_a(s, s') = \mathbb{P}[S_{t+1} = s' \mid S_t = s, A_t = a]$ → P predicts the next state

- $R_a(s) = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$ → R predicts the next (immediate) reward.

RL Agents.

- Value Based : Finding best value leads to best policy.

- Policy: Finding better policy through better value functions iteratively.

Learning Paradigm

- Model free
  - Policy / Value Function
- Model based
  - Policy / Value Function

Exploration and Exploitation.

- RL $\rightarrow$ a trail and error learning
- The agent's current policy should be able to discover a better policy if it exists.
- The agent current policy should be able to 'explore' different / better ways of doing the same task.

Exploration and Exploitation

- **Exploration** : Find more info about the environment.

Exploitation : Utilize the acquired info to obtain the best policy.

An agent should Explore & Exploit.

What is the best way ??

RL Agent Taxonomy